

# A simple way to speed up $qq \rightarrow 4\ell$ generation at NNLO

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We show that Machine Learning regressors can be used to infer the "slow" pieces of the  $qq \rightarrow 4\ell$  NNLO amplitude in  $\mathcal{O}(10^{-5})$  seconds (as opposed to the 10 seconds one requires to compute them exactly). This speed up can be achieved while keeping the errors on the NNLO correction below 10% and the error on the total NNLO amplitude below 0.1%. The speed up of the "slow" terms in the NNLO correction results in an increase of  $\mathcal{O}(10^4)$  in the NNLO correction to the  $qq \rightarrow 4\ell$  amplitude, bringing it down to the millisecond range. This improvement can be done relatively easily since there are many regions in the PS where the "slow" pieces are sub-leading and therefore one can infer those regions less precisely without a large impact on the full NNLO correction.

## INTRODUCTION

Need precision for ILC, FCC-ee and CLIC, due to high luminosity, hence need to go to N(N)LO. Problem with time consuming MC generation. Machine learning can solve this, both with phase space integration [] and more recently some works trying to make the evaluation of amplitudes to be faster [1, 2].

A lot of work relating diboson at high  $\sqrt{s}$  and the search for new physics [3–21]. **Motivate why we choose WZ from BSM most important channel**

In previous work [1] we showed that XGBoost can speed up the on loop  $gg \rightarrow ZZ$  amplitude squared by a factor  $\sim 10^3$  with errors in the double differential distribution below  $\lesssim 1\%$ . Here we continue that program by studying a more interesting and challenging process like  $qq \rightarrow VV \rightarrow 4\ell$ .

It is advantageous to separate the helicity amplitudes with intermediate, off-shell, vector bosons using the completeness relations for the polarization vectors into production and decays helicity amplitudes for three reasons:

1. this allows us to fit a 4d rather than 8d phase-space,
2. Naively, the  $2 \rightarrow 2$  helicity sub-amplitudes are well behaved function since one expects the point to point "j-factors" to be rather flat. This is more intuitive than opposition to the form factors  $A_i$ , can be much more intuitively and therefore, easily, normalized to reduce the error in the fit. We can fit flatter functions **(we should plot the ratios, aka the helicity amplitude "j-factors" for helamps, and also for the A's (in the Appendix / supplemental material))**. Nonetheless there could be better functions to fit, which we leave for future work.

### 3. Anything else?

We will discuss these three points in turn.

**The main goal of this work** is to speed up the evaluation of the "slow" 2-loop  $\times$  tree amplitudes in massless

QCD by using ML regressors to predict them. Notice that not all these amplitudes are "slow", since some of them are of the form (tree level amplitude  $\times$  form factor) where the form factor can be readily evaluated.

**How to be sure that there are no large cancellations between the "slow" terms we fit and the rest we don't when computing at least the total NNLO part of the amplitude.**

## SLOW 2-LOOP AMPLITUDES

The squared amplitude of the  $qq \rightarrow 4\ell$  process can be perturbatively expanded in  $\alpha_s$  at NNLO as follows

$$|\mathcal{M}_{2 \rightarrow 4}|^2 = |\mathcal{M}_{(0)}|^2 + \left(\frac{\alpha_s}{2\pi}\right) 2\Re[\mathcal{M}_{(0)}^* \mathcal{M}_{(1)}] + \left(\frac{\alpha_s}{2\pi}\right)^2 \left(|\mathcal{M}_{(1)}|^2 + 2\Re[\mathcal{M}_{(0)}^* \mathcal{M}_{(2)}]\right). \quad (2)$$

The NNLO pieces are given by the 1-loop amplitude squared  $|\mathcal{M}_{(1)}|^2$  and the interference term between the 2-loop and tree amplitudes  $2\Re[\mathcal{M}_{(0)}^* \mathcal{M}_{(2)}]$ . Given that **OpenLoops** can easily compute the  $|\mathcal{M}_{(1)}|^2$  term in milliseconds, we will focus on those terms for which each PS point requires  $t \lesssim \mathcal{O}(10)$  seconds to be computed. These correspond to the  $2\Re[\mathcal{M}_{(0)}^* \mathcal{M}_{(2)}]$  containing diagrams from the classes  $A, B, C$  shown in Fig. 2.

**Subtlety with same sign electrons**

## 2 $\rightarrow$ 2 SUB-AMPLITUDES

Given that all the  $qq \rightarrow 4\ell$  amplitudes we are interested in are double-resonant, i.e. of the form  $qq \rightarrow VV$ , with each  $V$  decaying to 2 leptons. We divide the full amplitude in  $2 \rightarrow 2$  helicity sub-amplitudes. We do this for various reasons. First of all our preliminary tests showed that we could achieve more precision by fitting the different sub-amplitudes which are functions of four variables than by fitting the full  $2 \rightarrow 4$  amplitude which depends

|                                          | $A$                                         | $B$                                        | $C$ | $F_W$                 | $F_Z$                 | $F_\gamma$ |
|------------------------------------------|---------------------------------------------|--------------------------------------------|-----|-----------------------|-----------------------|------------|
| $u_L \bar{d}_L \rightarrow W^+ Z$        | $\frac{-3 + 2e s_W^2}{6c_W s_W^2 \sqrt{2}}$ | $\frac{3 - 4e s_W^2}{6c_W s_W^2 \sqrt{2}}$ | 0   | $\frac{\pm c_W}{s_W}$ | 0                     | 0          |
| $u_L \bar{d}_L \rightarrow W^+ \gamma$   | $\frac{-e}{s_W 3\sqrt{2}}$                  | $X$                                        | 0   | $\frac{\pm c_W}{s_W}$ | 0                     | 0          |
| $q_L \bar{q}_L \rightarrow W^+ W^-$      | $X$                                         | $X$                                        | 0   | 0                     | $\frac{\pm c_W}{s_W}$ | $\pm 1$    |
| $q_R \bar{q}_R \rightarrow W^+ W^-$      | $X$                                         | $X$                                        | 0   | 0                     | $\frac{\pm c_W}{s_W}$ | $\pm 1$    |
| $q_L \bar{q}_L \rightarrow ZZ$           | $X$                                         | $X$                                        | 0   | 0                     | 0                     | 0          |
| $q_L \bar{q}_L \rightarrow Z\gamma$      | $X$                                         | $X$                                        | 0   | 0                     | 0                     | 0          |
| $q_L \bar{q}_L \rightarrow \gamma\gamma$ | $X$                                         | $X$                                        | 0   | 0                     | 0                     | 0          |
| $q_R \bar{q}_R \rightarrow ZZ$           | $X$                                         | $X$                                        | 0   | 0                     | 0                     | 0          |
| $q_R \bar{q}_R \rightarrow Z\gamma$      | $X$                                         | $X$                                        | 0   | 0                     | 0                     | 0          |
| $q_R \bar{q}_R \rightarrow \gamma\gamma$ | $X$                                         | $X$                                        | 0   | 0                     | 0                     | 0          |

Figure 1. couplings for blabla

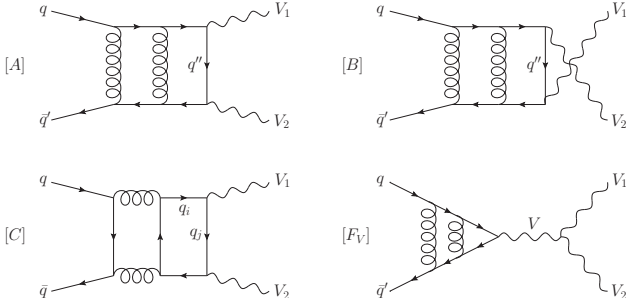


Figure 2. Representative diagrams contributing to  $qq \rightarrow 4\ell$ . Classes  $A$  and  $B$  present at tree, 1-loop and 2-loops, while class  $C$  only at 2-loops. Diagrams with class  $F$  topology are trivial to compute and therefore we do not consider in this work any terms containing them. Figure taken from [22].

on 8 variables. Therefore we rewrite the double-resonant  $qq \rightarrow VV \rightarrow 4\ell$  amplitudes as follows:

$$\begin{aligned} \mathcal{M}_{2 \rightarrow 4} &= \mathcal{M}_{2 \rightarrow 2}^{\mu\bar{\mu}} P_{\mu\nu} P_{\bar{\mu}\bar{\nu}} J^\nu J^{\bar{\nu}} \\ &= \sum_{\kappa\lambda, X} [(\mathcal{M}_{2 \rightarrow 2}^X)^{\mu\bar{\mu}} (\epsilon_\mu^\kappa)^* (\epsilon_{\bar{\mu}}^\lambda)^*] (\epsilon_\nu^\kappa J^\nu) (\epsilon_{\bar{\nu}}^\lambda J^{\bar{\nu}}) \Delta_3 \Delta_4 \end{aligned} \quad (3)$$

where in the first line  $\mathcal{M}_{2 \rightarrow 2}^{\mu\bar{\mu}}$  is the  $qq \rightarrow V_3 V_4$  amplitude,  $P_{\mu\nu}$ ,  $P_{\bar{\mu}\bar{\nu}}$  are the gauge boson propagators and  $J^\nu$ ,  $J^{\bar{\nu}}$  are the lepton currents. In the second line  $(\mathcal{M}_{2 \rightarrow 2}^X)$  is the  $2 \rightarrow 2$  amplitude for each of the classes  $X = A, B, C, F_V$  shown in Fig. 2,  $\Delta_i = (p_i^2 - M_i^2)^{-1}$  while  $\epsilon_\mu^\lambda$ ,  $\epsilon_{\bar{\mu}}^\kappa$  are the intermediate gauge boson polarizations with  $\lambda, \kappa = \pm, 0$ .

Note that there are large cancellations between dia-

grams of different classes. The result of their sum is orders of magnitude smaller than the size of each individual diagram. This is the case between classes  $A$  and  $B$  for  $ZZ/Z\gamma/\gamma\gamma$  and  $A, F$  (or  $B, F$ ) for  $WW$  and  $WZ/W\gamma$ . This fact forbids us to fit each class individually once, and then just re-scale them with appropriate couplings for each process. Once this advantage is lost we then proceed to train together all the class of diagrams together for each final state. For each  $qq \rightarrow VV'$  helicity one should train the following combinations of amplitudes

$$\mathcal{M}_{qq'}^{[P]} = \sum_X g_{qq'}^{[P]} \mathcal{M}^{[X]} \quad (4)$$

where  $X \in A, B, C, F_V$ ,  $\mathcal{M}^{[X]}$  are the  $2 \rightarrow 2$  helicity amplitudes, and  $g_{qq'}^{[P]}$  are given in the following table:

## SYMMETRY PROPERTIES

The  $q_r \bar{q}_{\bar{r}} \rightarrow V_1 V_2$  helicity amplitudes  $\mathcal{M}_{\kappa\lambda}^{[X_r]}$  for classes  $A, B$  and  $C$  have certain symmetry relations under the transformations:

$$\begin{aligned} \pi_{12} &:= p_1 \leftrightarrow p_2 \Rightarrow \{t \leftrightarrow u\} \\ \pi_{34} &:= p_3 \leftrightarrow p_4 \Rightarrow \{t \leftrightarrow u, \quad p_3^2 \leftrightarrow p_4^2\}. \end{aligned} \quad (5)$$

These are the following

$$\begin{aligned}
\mathcal{M}_{\kappa\lambda}^{[A-]} &\xrightarrow{\pi_{12}} (-1)^{N_0} \mathcal{M}_{\kappa\lambda}^{[B+]}, \\
\mathcal{M}_{\kappa\lambda}^{[A-]} &\xrightarrow{\pi_{34}} -\mathcal{M}_{\lambda\kappa}^{[B-]}, \\
\mathcal{M}_{\kappa\lambda}^{[A-]} &\xrightarrow{\pi_{12} \circ \pi_{34}} (-1)^{N_0+1} \mathcal{M}_{\lambda\kappa}^{[A+]}, \\
\mathcal{M}_{\kappa\lambda}^{[A(B)-]} &= -\mathcal{M}_{\bar{\kappa}\bar{\lambda}}^{[A(B)+]}, \\
\mathcal{M}_{\kappa\lambda}^{[C-]} &\xrightarrow{\pi_{12}} (-1)^{N_0} \mathcal{M}_{\kappa\lambda}^{[C+]}, \\
\mathcal{M}_{\kappa\lambda}^{[C+]} &= \mathcal{M}_{\bar{\kappa}\bar{\lambda}}^{[C-]}, \\
\mathcal{M}_{\kappa\lambda}^{[C-]} &\xrightarrow{\pi_{34}} -\mathcal{M}_{\lambda\kappa}^{[C-]}, \\
\text{\textcolor{red}{class F?}}
\end{aligned} \tag{6}$$

where the labels  $A_{\pm}$  and  $B_{\pm}$  indicate the class and handedness of the helicity amplitudes (with the sub-indices  $+, -$  denoting the right and left-handed amplitudes respectively i.e. the eigenvalues of  $\gamma_5$ ). The indices  $\kappa, \lambda$  are the polarizations of  $V_1 V_2$ . In the expressions,  $\bar{\kappa}, \bar{\lambda}$  correspond to flipping the helicity from  $+$  to  $-$  and viceversa (while there is no change if  $\bar{\kappa} = 0$  or  $\bar{\lambda} = 0$ ) while  $N_0$  is the number of longitudinal polarizations  $\kappa, \lambda$  that equal zero.

The following transformation are combinations of the previous ones in (A.1)-(A.4), and are useful when inferring the helicity amplitudes fitting the least number of regressors as explained in [Sec. X](#).

$$\mathcal{M}_{\kappa\lambda}^{[A-]} \xrightarrow{\pi_{12}} (-1)^{N_0+1} \mathcal{M}_{\bar{\kappa}\bar{\lambda}}^{[B-]} \tag{7}$$

$$\mathcal{M}_{\kappa\lambda}^{[A-]} \xrightarrow{\pi_{12} \circ \pi_{34}} (-1)^{N_0} \mathcal{M}_{\bar{\lambda}\bar{\kappa}}^{[A-]} \tag{8}$$

[Need to write the symmetry properties that we used \(the useful ones\) for each of the combinations that we say we can fit](#)

WW:  
WZ/ $\gamma\gamma$ :  
ZZ/ $\gamma\gamma$ :  
Z $\gamma$ :

## RESULTS

In this section we show ...

## SUMMARY AND CONCLUSIONS

We have shown that ML regressors...

There are various other functions that one could try to fit, and this we will do in the future.

Also we plan to implement these ML regressors in MATRIX ??.

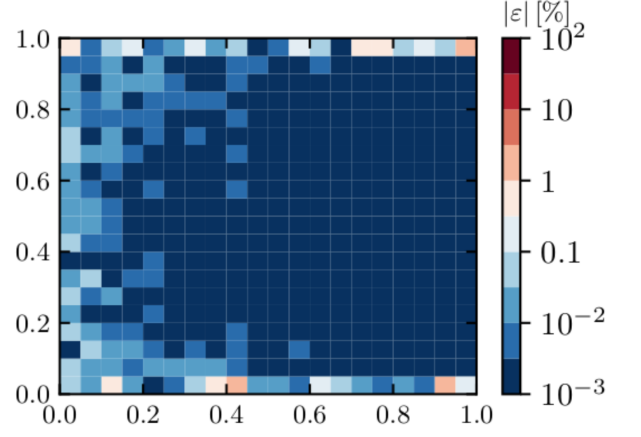


Figure 3. bla bla [put correct plot](#)

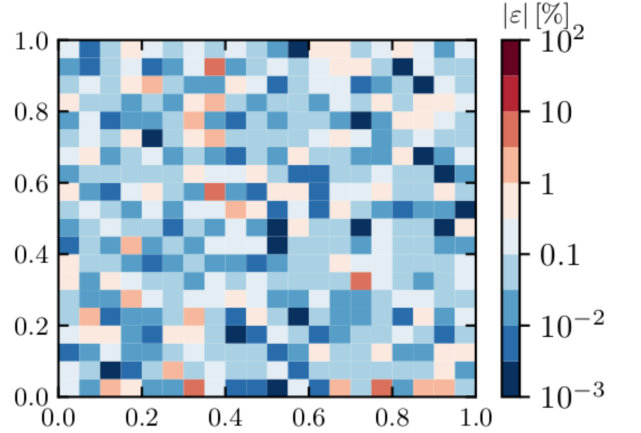


Figure 4. bla bla [put correct plot](#)

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